

# Economic Denial of Sustainability (EDoS) Mitigation Approaches in Cloud: Analysis and Open Challenges

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**Abstract-** Economic denial of sustainability (EDoS) appeared to be a new menace of cloud computing. This pristine attack is a breed of DoS or DDoS attack that targets the vulnerabilities of cloud consumers utility pricing model. EDoS attackers steadily send illegitimate traffic to gradually consume cloud resources such as virtual machines, virtual network devices, virtual security devices and databases so that it can trigger auto scaling features of cloud. Consequently, targeted cloud consumer ends up with an exorbitant amount of bill that may lead to bankruptcy. The other side effect of this attack is the persistent degradation of services faced by benign cloud users. This paper gives an overview of mitigation approaches proposed by researchers over the years exclusively for EDoS attack defense. We present a taxonomy of EDoS mitigation strategies with circumspect. The taxonomy focus evaluation metric used to mitigate EDoS along with its applicability in cloud environment. We further explore some open challenges that are yet to come into the light. Finally, a probable future direction is given at the end of this paper.

**Keywords-** *Economic Denial of Sustainability; EDoS; Cloud Computing; Mitigation, Approaches; Auto Scaling; False Rate; Response Time; DDoS.*

## I. INTRODUCTION

Cloud Computing offers some of the exclusive features such as pay according to usage, elasticity, auto scalability that are considered an opportune demand over traditional computing services. Organizations are getting involved in cloud services to minimize their capital expenditure (Capex). Moreover, small organizations who couldn't afford bearing their own ICT infrastructural cost are found cloud services more attractive. However, security in cloud computing deemed as the primary barrier. Distributed Denial of Service (DDoS), Botnet, Co-resident or Side channel attacks are widely regarded as major threats that can affect cloud environment. Recently, security experts and researchers revealed a new type of attack which fraudulently consumes the resources of cloud by exploiting two exciting features of cloud, i) Pay per use and ii) On demand elasticity. As a result,

cloud consumers resource utility bill goes up significantly without any justification. Such a stealthy nature of attack that emerges from DDoS but with a different malicious objective is known as Economic Denial of Sustainability (EDoS) [1].

Since EDoS is relatively new form of attack and due to its tricky nature, researchers and security experts are hardly paid attention to analyze this attack over the years. Another reason is the misconception of EDoS among researchers. Some are getting confused by believing that DDoS and EDoS attack are same. Nonetheless, some researchers exclusively conducted their study on EDoS attack, its impact and its mitigation. In this work, we organize and critically analyze the papers based on the EDoS attack mitigation approaches proposed by the academicians over the years. Our goal is to point out issues regarding EDoS attack mitigations, their challenges and future direction.

During analyzing the existing solutions of EDoS attack, we focus on the mitigation strategy, methodology, performance metrics used in each literature. Further, we address the pros and cons of those techniques in terms of applicability, feasibility and adaptability in cloud environment. Limitations and scope of each approach are addressed in our paper. We also discuss about some open issues and challenges regarding EDoS mitigation.

The remaining parts of our work is arranged as follows. In section II, we discuss about EDoS attack and its impact. Section III presents, taxonomy of existing EDoS mitigation approaches. Discussion and challenges of EDoS mitigation are described in section IV. Finally, conclusion with future direction is depicted in section V.

## II. EDOS ATTACK AND ITS IMPACT IN CLOUD

The term Economic Denial of Sustainability (EDoS) is first referred by Christofer Hoff in 2008 [1]. In EDoS attack, attacker intelligently and intentionally maneuvers the auto scaling feature in cloud environment so that computing resources of a targeted cloud consumer scales up unexpectedly. As a result, cost associated with cloud resource provisioning affects the utility pricing to the targeted cloud consumer [2]. This persistent effort over a period ends up with an exceedingly high amount of bill to the cloud consumer.

Thus, economic feasibility of the cloud consumer becomes unsustainable. Fig. 1 depicts such scenario as follows.

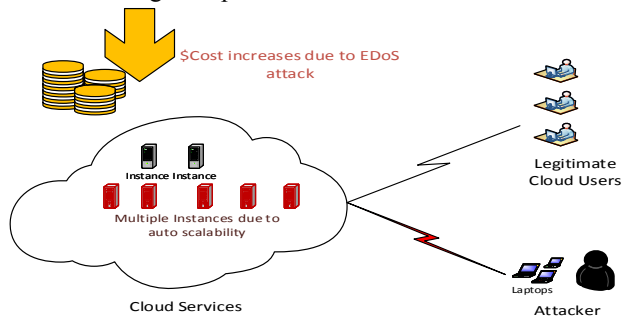


Fig. 1. Billing effect of cloud consumer due to EDoS attack

In DDoS attack, the target of the attacker is to interrupt the services offered by cloud service provider. So, DDoS attackers in most of the cases, irrationally launch attack with their maximum resources over a short span of time. The goal of such attack is to entirely shut down the service or significantly degrade the quality of service. Conversely, in EDoS attack, attackers usually seem more smart and rational. They target an individual or specific group of cloud consumer like a spy. They slowly and gradually push illegitimate traffic over a longer period. The motive is to maximize the financial cost incurred by the targeted cloud consumer through over provisioning of cloud resources. Traffic pattern of EDoS is somewhat different compared to the DDoS traffic pattern as shown in fig. 2. It is stealthy in nature and remains below the threshold value of typical intrusion detection system so that it can pass the filter like legitimate traffic.

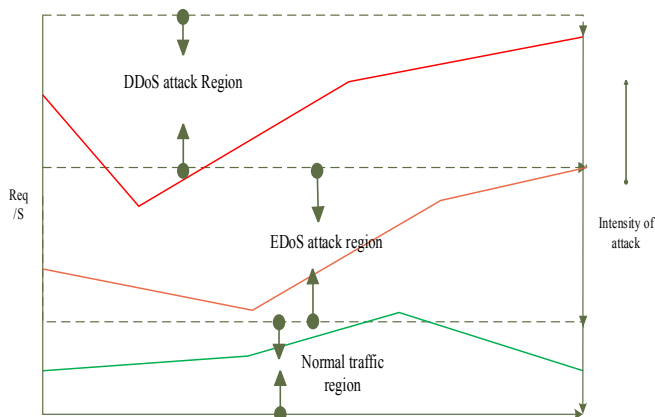


Fig. 2. EDoS attack region with attack strength

EDoS attack affects the cloud computing steadily both in performance and pricing model. EDoS in the form of fraudulent resource consumption can also cause a significant additional cost of energy bill. Ficco et al. [3] stated that if the attack involved 1000 application servers in Amazon EC2, it consumes an additional 1440 kWh power in a day cost 5184\$/month based on US current electricity market price.

Haidari et al. [4] evaluated the impact of EDoS attack in cloud services. Their [4] experimental outcome conclude that EDoS attack has significant impact on both the performance and the billing of cloud services. The result shows, an EDoS attack at a rate of 6000 req/s targeting VM of a cloud consumer cost 15 times more than the cost of its normal usage. Some other researchers are also investigated on EDoS attack in cloud and explored their potential impact [5-11]. In this paper, our focus is on analysis of mitigation approaches proposed by the researchers. So, further impact analysis of EDoS by various researchers is out of the scope of this work.

### III. EDoS ATTACK MITIGATION APPROACHES – A TAXONOMY

Several researchers proposed their model to alleviate EDoS attack. In this section, we illustrate existing mitigation tactics in table 1 by highlighting model, strategy and evaluation metrics used in each approach. Each of the model is analyzed critically with an insight to obtain the gap which further explore the open challenges.

1. **sPoW:** Self verifying proof of work (sPoW) is a preliminary approach to mitigate EDoS. Khor and Nakao [12] designed sPoW in such a way that it can be easily deployable into the existing cloud technology. sPow acts like a DNS server but it returns crypto puzzle with an encryption key instead of IP address. In this model, client communicate with the server through plugins embedded in a static webpage. The server plugin handles the puzzle generation by creating a temporary encryption channel and sends back to the client as a puzzle request to solve. Upon successful accomplishment of solving puzzle, the server ensures a secure communication channel for information exchange between client and itself. For an unsuccessful attempt, it requests for more difficult crypto-puzzle to solve with a view to re-establish the connection through a higher priority channel. The proposed scheme is not evaluated in terms of any metric to validate the model. There are some limitations of the scheme that are even addressed by the authors themselves. Firstly, attacker can easily launch a puzzle accumulation attack without solving crypto-puzzle based request. Secondly, the scheme is vulnerable to false positive rate as some legitimate users might be denied of getting service due to puzzle difficulty. Moreover, crypto-puzzled based solver attributes more computational cost that might cause the delay to the response time. Nonetheless, the scheme still has its acceptance due to its well-suited adaptability in the cloud environment.

TABLE 1. CLASSIFICATION OF EXISTING EDoS MITIGATION TECHNIQUES

| Approach                                  | Model                                                 | Mitigation Strategy                                                                   | Evaluation Metric |               |                 |      |
|-------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------|---------------|-----------------|------|
|                                           |                                                       |                                                                                       | False Rate (FR)   | Response Time | CPU Utilization | Cost |
| <b>Khor &amp; Nakao, 2009 [12]</b>        | Self-verifying proof of work (sPoW)                   | Used Crypto-Puzzle based solver through client-server plugins                         | x                 | x             | x               | x    |
| <b>Idziorek &amp; Tannian, 2011[13]</b>   | Cloud Utility Model for Profit & Ruin                 | a) Zipf's Law<br>b) Entropy Detection                                                 | ✓                 | x             | x               | ✓    |
| <b>Sqalli et al., 2011, 2016 [14, 15]</b> | EDoS-Shield                                           | Packet filtering based on Graphical Turing Test                                       | x                 | ✓             | x               | ✓    |
| <b>Haidari et al., 2012 [16]</b>          | Enhanced EDoS-Shield                                  | Incorporate time to live (TTL) value in EDoS - Shield                                 | x                 | ✓             | ✓               | ✓    |
| <b>Kumar et al., 2012 [17]</b>            | In-Cloud Scrubber Service                             | Crypto-Puzzle solver used in reactive manner                                          | x                 | x             | x               | x    |
| <b>Masood et al., 2013 [18]</b>           | EDoS Armor                                            | a) Authentication<br>b) Admission Control<br>c) Congestion Control                    | x                 | ✓             | ✓               | x    |
| <b>Koduru et al., 2013 [19]</b>           | Time Spent on Web Page (TSP)                          | Detect attack based on time spent on web page                                         | x                 | ✓             | x               | x    |
| <b>Baig et al., 2013, 2016 [2, 20]</b>    | Controlled access to cloud resources                  | Reactive approach based rate limiting technique                                       | ✓                 | ✓             | ✓               | x    |
| <b>Alosaimi et al., 2015 [21]</b>         | Enhanced DDoS Mitigation System                       | Proactive approach using client puzzle, IPS & reverse proxy                           | x                 | ✓             | ✓               | x    |
| <b>Thaper &amp; Verma, 2015 [22]</b>      | Adaptive Pattern Attack Recognition Technique (APART) | Anomaly based detection based on traffic pattern                                      | x                 | ✓             | ✓               | x    |
| <b>Haidari et al. , 2016 [23]</b>         | Analytical Modeling of EDoS-Shield                    | Used queuing modeling to analyze the performance of EDoS-Shield                       | x                 | ✓             | ✓               | x    |
| <b>Karami &amp; Chen, 2016 [24]</b>       | Markov-based Model                                    | Anomaly based detection using<br>1. Markov Chain Model<br>2. Hidden Semi-Markov Model | ✓                 | x             | x               | x    |
| <b>Chowdhury et al., 2017 [25]</b>        | EDoS Eye                                              | Optimal threshold setting using game theory                                           | x                 | x             | x               | x    |

2. **Idziorek & Tannian Model:** The authors [13] mainly focus on the vulnerability of current cloud utility pricing model. It shows how EDoS attack in the form of fraudulent resource consumption gradually affects the cost model. They used pricing model of Amazon EC2 and measured the cost due to aggregate EDoS attack for a specific period. To countermeasure, they proposed two detection methodologies- i) Zipf's Law and ii) Entropy detection. Zipf's Law is used to detect anomalous pattern of traffic. On the other hand, entropy based detection analyzes individual user traffic nature in an entropy of session lengths over a specific time. In both cases, they evaluated false positive rate and false negative rate independently.
3. **EDoS-Shield:** This model is first proposed by [14]. After that, several researchers are conducted their works based on EDoS-Shield [15, 16, 23]. The architecture of EDoS-Shield consists of virtual firewalls (VFs) and verifier nodes (V-nodes) as shown in fig. 3. below

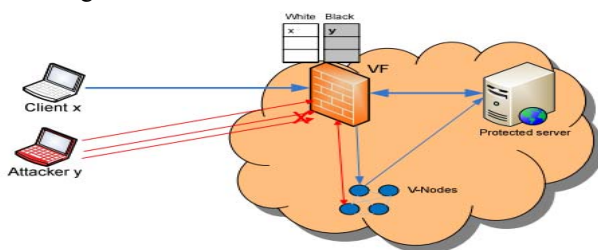


Fig. 3. Architecture of EDoS-Shield [14]

In EDoS-Shield, clients request is received by virtual firewall which forwards the request to V-node. V-node, in reply sends graphical Turing test to the client. If the client passes the test, her requests is direct to the server and the source of her IP address is updated to the whitelist in VF. In contrast, failing the test leads to source IP address into the blacklist of VF while subsequent packet request is dropped. This model is quite effective to prevent botnet generated traffic. However, this model has few shortcomings. It cannot filter malicious traffic if the source IP address is spoofed. [16] overcome this limitation by incorporating time to live (TTL) value into each IP address in their proposed Enhanced EDoS-Shield Model. Yet, false rate which is considered a crucial factor is not been evaluated in this model. In addition, use of graphical Turing test though good for verification purpose but not good enough for disable people. Moreover, graphical Turing test and updating IP lists (white & black) in VF increase end to end latency. EDoS-Shield is further experimented in a testbed [15] to compare experimental results of CPU utilization and response time with their previous simulation results. [23] performed an analytical

modeling of EDoS-Shield using queuing theory but false rate evaluation remains unaddressed.

4. **In-Cloud Scrubber Service:** In this model [17], service operates in two modes depend on server load and bandwidth consumption. These are i) Normal Mode and ii) Suspected Mode. When both the server resource and bandwidth traffic exceed their corresponding threshold values then service provider switches to suspect mode expecting a high-volume of DDos attack. In that case, to avail the on-demand service client must solve difficult puzzle. Conversely, if server resources exceed the threshold value but bandwidth usage are below its threshold value then a moderate puzzle solve request send to client. Since no evaluation has been done such as false rate, CPU utilization, response time and cost in the proposed model, applicability of this model in cloud remains doubtful.
5. **EDoS Armor:** The proposed technique [18] is designed for e-commerce application. The mechanism has two phase defense system, consists of admission control and congestion control. During admission control each new client can only be accessed by solving either image based challenge or cryptographic challenge. The purpose of admission control is to limit the number of end users. Congestion control set priorities to the client based on their past browsing behavior which classified as good and bad user. This model is complicated from the user perspective. The adaptability of this model in any e-commerce site come into question when potential new clients might lose interest using the site because of its too much complexity in nature.
6. **Time Spent on Web Page:** [19] presented a technique based on time spent on web page. The main contribution of this paper is to detect the attack traffic by observing the deviation from mean time spent on a web page (TSP). The authors of the scheme used mean absolute deviation to measure and compare EDoS traffic with legitimate traffic. However, the scheme to some extent, can be useful to identify botnet attack but the responsive action or mitigation approach is not addressed. Hence, this scheme can only be applied for EDoS detection but not for any protection or prevention.
7. **Controlled access to cloud resources:** The scheme [2] mitigates EDoS in a reactive manner. The architecture is somewhat similar to EDoS-Shield comprised of vFirewall and VM Investigator. vFirewall filters the traffic based on Turing test and

updated blacklist. VM investigator probe the suspected request to further analysis. Unlike EDoS-Shield, this scheme allows traffic that fail the Turing test and direct to VM Investigator for deep inspection, where the former dropped the traffic. The scheme uses a threshold parameter based on CPU usage metric. This threshold parameter (upper & lower bound) helps to trigger auto scaling feature in cloud. Although, auto scaling not solely depends on threshold value. Rather, a specific observation time of anomalous resources is also considered along with threshold to enable auto scaling. The scheme also uses a parameter name User Trust Factor (UTF) to evaluate end user's values based on pass or fail the Turing test. This scheme is among the few mitigation approaches which evaluated the false rate. It also evaluated latency and CPU usage.

8. **Enhanced DDoS-MS:** This mitigation framework [21] consists of a firewall, a verifier node (VN) , a puzzle server, an IPS and a reverse proxy placed in front of the protected zone of VMs. The scheme emphasizes the capability of firewall which controls the overall protection process. VN conducts the graphical Turing test and reports to the firewall. IPS inspects the packet to detect the malicious activities. Reverse proxy (RP) is used to hiding target server, controlling load balancer and monitoring the traffic rate. Further, client puzzle server is used to limit the activities of suspicious users detected in RP. The increased complexity of design made it impractical to adopt this model in cloud environment. In addition, several authentication processes intensify the latency significantly.
9. **APART:** Adaptive Pattern Attack Recognition Technique (APART) [22] is an anomaly based detection technique based on traffic pattern recognition. The algorithm is designed such a way that it first scans the traffic flow. Then, it checks the traffic for any abnormalities. If found, then it compares with predefined threshold value of traffic rate and determine whether to allow this traffic through firewall or not. The authors of that scheme neither make a clear explanation of their proposed algorithm nor elaborate functionality of their model. Moreover, they illustrate Baig et al.,2013 [20] architecture as their proposed architecture.
10. **Markov-based Model:** This scheme [24] is actually an anomaly based detection system. The authors work mainly focused to identify EDoS traffic rather than mitigation. The evaluation of the performance of

detection is done using two different Markov models. The first one is a simple Markov chain model and the second one is hidden semi-Markov model. The later model proved to be effective of detecting low rate attack traffic with some additional computational cost. Both false positive and false negative rate have been evaluated separately for these Markov models. The proposed EDoS attribution technique solely relies on footprint of resource usage by the users.

11. **EDoS Eye:** This is our mitigation model [25] applying game theory. To aim an effective mitigation approach, we consider attacker intents by modeling two player non-cooperative zero-sum static game. We obtained optimal threshold value through Nash equilibrium which helps reducing EDoS attacker's payoff considerably. Our model incorporate honeypot to minimize false rate. Further, as part of the extension work of EDoS Eye, we are in the experimental stage for evaluating response time and system throughput till the writing of this paper.

#### IV. DISCUSSION AND EDoS MITIGATION CHALLENGES

A taxonomy of mitigation approaches in the previous chapter reveals that most of the existing solutions [2, 12, 14-18, 20-23] are either use Turing test or crypto-puzzle based challenge server. This implies, researchers give priority to botnet attack while designing their proposed scheme so that it can recognize human generated traffic and machine generated traffic separately. Table 1 illustrates that majority of researchers in their proposed models evaluated response time and CPU utilization as performance metrics. Few researchers [2, 13, 15, 16, 24] are evaluated either false rate and/or cost. There are two things that need to be considered for designing an effective mitigation model. First one is the quality of service (QoS) and another is cloud utility pricing model. False rate, response time are associated with QoS whereas cost and CPU usage are associated with utility pricing model. Setting a threshold value is very important factor for defensive mechanism. In case of QoS in defensive mechanism, threshold value used to control traffic and prevent malicious traffic. This approach is a preventive approach. In reactive approach, threshold value can be used to trigger auto scaling or provisioning/deprovisioning of cloud resources. This approach is more suited for protection of cloud utility pricing model. Another important thing that also needs to be considered is to ensure QoS in the interim period.i.e. the period between attack and resolution.

The attackers while launching EDoS are very strategic. Unlike DDoS, EDoS traffic patterns are sophisticated in nature with more time span. Existing approaches used predefined threshold value that might lead to higher false rate. For an

efficient defensive model, setting an optimal threshold value though challenging but necessary. Analyzing attacker behavior or profile is also a key factor for proactive defensive mechanism. Most of the existing mitigation models overlooked this fact. Moreover, use of graphical Turing test or crypto-puzzle increased time complexity. Finding an alternative approach to optimize the response time is another open issue that needs to be resolved.

## V. CONCLUSION AND FUTURE DIRECTION

We painstakingly summarized and reviewed academic literature based on EDoS attack mitigation approaches. Selection of each paper is chosen based on thorough understanding. The taxonomy presented on the basis of each defensive model by focusing their evaluation metrics and adaptably in the cloud environment. We, critically analyzed the limitations of each selected approach with justification. Our meticulous effort on EDoS mitigation literatures explore some challenges that may lead to further research on that field with different approach. Understanding of EDoS attacker behavior, identification of their strategy, and proactive design of EDoS defensive mechanism are crucial for future research work. Use of mathematical modeling such as queuing model, statistical and probability analysis, game theory can be used to overcome existing challenges. Use of honeypot or dynamic firewall based algorithm to investigate traffic pattern instead of time intensive Graphical Turing test or Crypto-puzzle may improve the end to end response time.

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